

Chapter 9

Neural networks

9.1 Introduction

An *Artificial Neural Network* (ANN) models the relationship between a set of input signals and an output signal using a model derived from our understanding of how a biological brain responds to stimuli from sensory inputs. Just as a brain uses a network of interconnected cells called neurons to create a massive parallel processor, ANN uses a network of artificial neurons or nodes to solve learning problems.

9.2 Biological motivation

Let us examine how a biological neuron functions. Figure 9.2 gives a schematic representation of the functioning of a biological neuron.

In the cell, the incoming signals are received by the cell's *dendrites* through a biochemical process. The process allows the impulse to be weighted according to its relative importance or frequency. As the cell body begins accumulating the incoming signals, a threshold is reached at which the cell fires and the output signal is transmitted via an electrochemical process down the *axon*. At the axon's terminals, the electric signal is again processed as a chemical signal to be passed to the neighboring neurons across a tiny gap known as a *synapse*.¹

Biological learning systems are built of very complex webs of interconnected neurons. The human brain has an interconnected network of approximately 10^{11} neurons, each connected, on an average, to 10^4 other neurons. Even though the neuron switching speeds are much slower than

¹Neuron. (2018, February 15). In Wikipedia, The Free Encyclopedia. Retrieved 01:44, February 23, 2018.

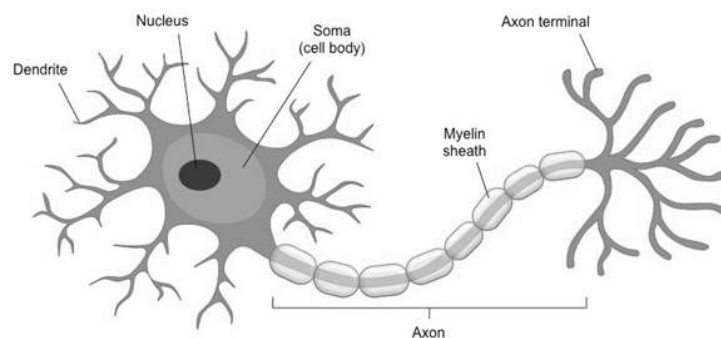


Figure 9.1: Anatomy of a neuron

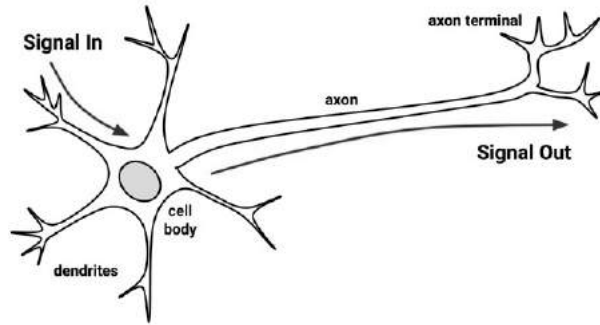


Figure 9.2: Flow of signals in a biological neuron

computer switching speeds, we are able to take complex decisions relatively quickly. Because of this, it is believed that the information processing capabilities of biological neural systems is a consequence of the ability of such systems to carry out a huge number of parallel processes distributed over many neurons. The developments in ANN systems are motivated by the desire to implement this kind of highly parallel computation using distributed representations.

9.3 Artificial neurons

Definition

An *artificial neuron* is a mathematical function conceived as a model of biological neurons. Artificial neurons are elementary units in an artificial neural network. The artificial neuron receives one or more inputs (representing excitatory postsynaptic potentials and inhibitory postsynaptic potentials at neural dendrites) and sums them to produce an output. Each input is separately weighted, and the sum is passed through a function known as an *activation function* or *transfer function*.

Schematic representation of an artificial neuron

The diagram shown in Figure ?? gives a schematic representation of a model of an artificial neuron. The notations in the diagram have the following meanings:

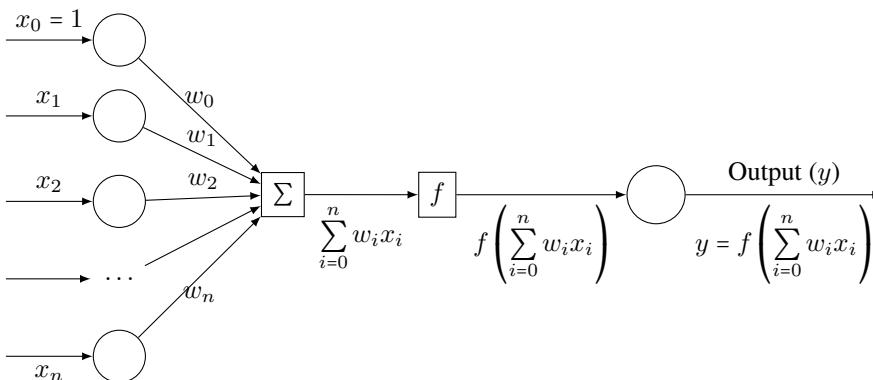


Figure 9.3: Schematic representation of an artificial neuron

$x_1, x_2, \dots, x_n$: input signals

$w_1, w_2, \dots, w_n$: weights associated with input signals

x_0 : input signal taking the constant value 1
 w_0 : weight associated with x_0 (called bias)
 $\sum$: indicates summation of input signals
 f : function which produces the output
 y : output signal

The function f can be expressed in the following form:

$$y = f\left(\sum_{i=0}^n w_i x_i\right) \quad (9.1)$$

Remarks

The small circles in the schematic representation of the artificial neuron shown in Figure 9.3 are called the *nodes* of the neuron. The circles on the left side which receives the values of $x_0, x_1, \dots, x_n$ are called the *input nodes* and the circle on the right side which outputs the value of y is called *output node*. The squares represent the processes that are taking place before the result is outputted. They need not be explicitly shown in the schematic representation. Figure 9.4 shows a simplified representation of an artificial neuron.

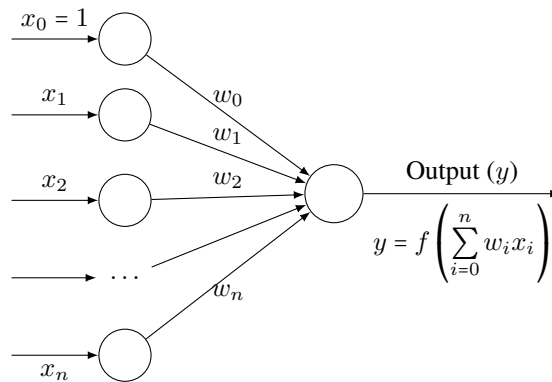


Figure 9.4: Simplified representation of an artificial neuron

9.4 Activation function

9.4.1 Definition

In an artificial neural network, the function which takes the incoming signals as input and produces the output signal is known as the *activation function*.

Remark

Eq.(9.1) represents the activation function of the ANN model shown in Figure ??.

9.4.2 Some simple activation functions

The following are some of the simple activation functions.

1. Threshold activation function

The *threshold activation function* is defined by

$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x \leq 0 \end{cases}$$

The graph of this function is shown in Figure 9.5.

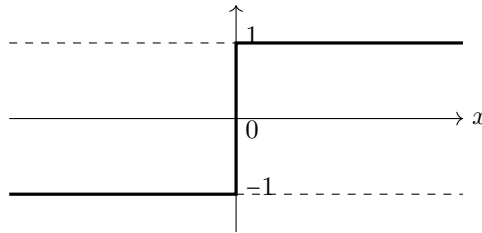


Figure 9.5: Threshold activation function

2. Unit step functions

Sometimes, the threshold activation function is also defined as a unit step function in which case it is called a *unit-step activation function*. This is defined as follows:

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

The graph of this function is shown in Figure 9.6.

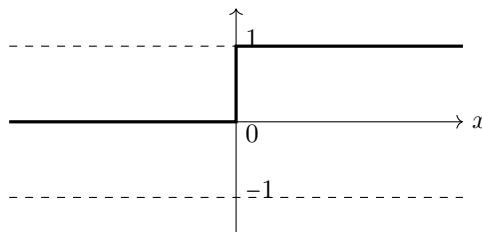


Figure 9.6: Unit step activation function

3. Sigmoid activation function (logistic function)

One of the most commonly used activation functions is the sigmoid activation function. It is defined as follows:

$$f(x) = \frac{1}{1 + e^{-x}}$$

The graph of the function is shown in Figure 9.7.

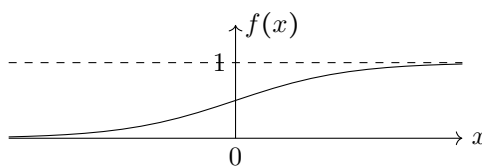


Figure 9.7: The sigmoid activation function

4. Linear activation function

The linear activation function is defined by

$$F(x) = mx + c.$$

This defines a straight line in the xy -plane.

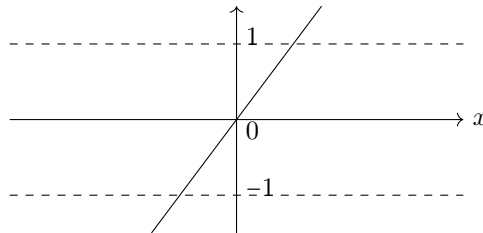


Figure 9.8: Linear activation function

5. Piecewise (or, saturated) linear activation function

This is defined by

$$f(x) = \begin{cases} 0 & \text{if } x < x_{\min} \\ mx + c & \text{if } x_{\min} \leq x \leq x_{\max} \\ 0 & \text{if } x > x_{\max} \end{cases}$$

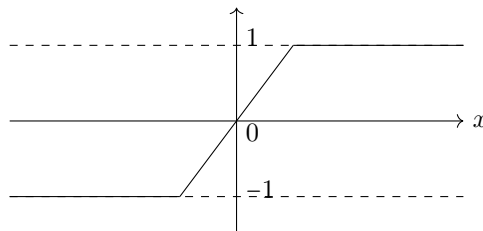


Figure 9.9: Piecewise linear activation function

6. Gaussian activation function

This is defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

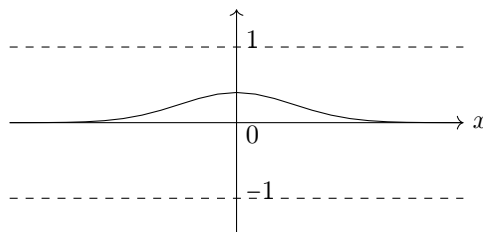


Figure 9.10: Gaussian activation function

7. Hyperbolic tangential activation function

This is defined by

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

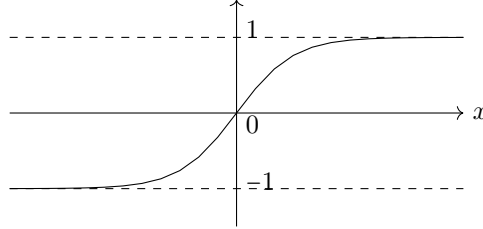


Figure 9.11: Hyperbolic tangent activation function

9.5 Perceptron

The perceptron is a special type of artificial neuron in which the activation function has a special form.

9.5.1 Definition

A perceptron is an artificial neuron in which the activation function is the threshold function.

Consider an artificial neuron having $x_1, x_2, \dots, x_n$ as the input signals and $w_1, w_2, \dots, w_n$ as the associated weights. Let w_0 be some constant. The neuron is called a perceptron if the output of the neuron is given by the following function:

$$o(x_1, x_2, \dots, x_n) = \begin{cases} 1 & \text{if } w_0 + w_1x_1 + \dots + w_nx_n > 0 \\ -1 & \text{if } w_0 + w_1x_1 + \dots + w_nx_n \leq 0 \end{cases}$$

Figure 9.12 shows the schematic representation of a perceptron.

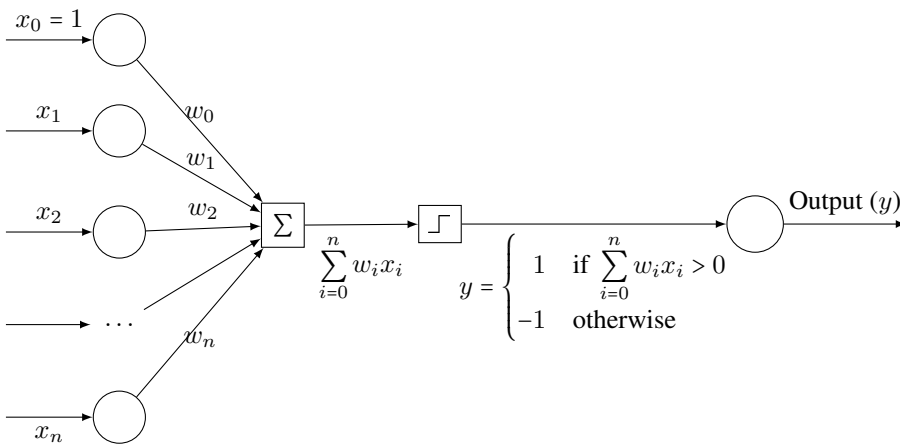


Figure 9.12: Schematic representation of a perceptron

Remarks

1. The quantity $-w_0$ can be looked upon as a “threshold” that should be crossed by the weighted sum $w_1x_1 + \dots + w_nx_n$ in order for the neuron to output a “1”.

9.5.2 Representations of boolean functions by perceptrons

In this section we examine whether simple boolean functions like x_1 AND x_2 can be represented by perceptrons. To be consistent with the conventions in the definition of a perceptron we assume that the values -1 and 1 represent the boolean constants “false” and “true” respectively.

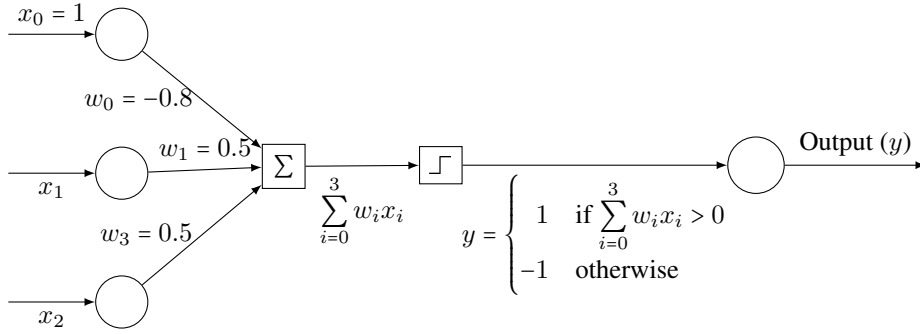
9.5.3 Representation of x_1 AND x_2

Let x_1 and x_2 be two boolean variables. Then the boolean function x_1 AND x_2 is represented by Table 9.1. It can be easily verified that the perceptron shown in Figure 9.13 represents the function

x_1	x_2	x_1 AND x_2
-1	-1	-1
-1	1	-1
1	-1	-1
1	1	1

Table 9.1: The boolean function x_1 AND x_2

x_1 AND x_2 .

Figure 9.13: Representation of x_1 AND x_2 by a perceptron

In the perceptron shown in Figure 9.13, the output is given by

$$\begin{aligned}
 y &= \begin{cases} 1 & \text{if } \sum_{i=0}^3 w_i x_i > 0 \\ -1 & \text{otherwise} \end{cases} \\
 &= \begin{cases} 1 & \text{if } -0.8 + 0.5x_1 + 0.5x_2 > 0 \\ -1 & \text{otherwise} \end{cases}
 \end{aligned}$$

Representations of OR, NAND and NOR

The functions x_1 OR x_2 , x_1 NAND x_2 and x_1 NOR x_2 can also be represented by perceptrons. Table 9.2 shows the values to be assigned to the weights w_0, w_1, w_2 for getting these boolean functions.